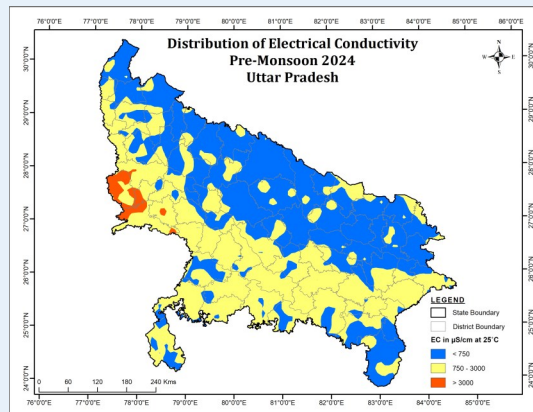


Groundwater Quality Scenario in Uttar Pradesh

Parameters	No of samples	Permissible limit	No of Samples above permissible limit	% Samples above permissible Limit
EC	1333	3000 $\mu\text{S/cm}$	31	2.33
Fluoride	1333	1.5 mg/L	54	4.05
Nitrate	1333	45 mg/L	107	8.03
Arsenic	612	10 ppb	23	3.76
Uranium	612	30 ppb	29	4.74



Districts with anomalous values at sporadic locations

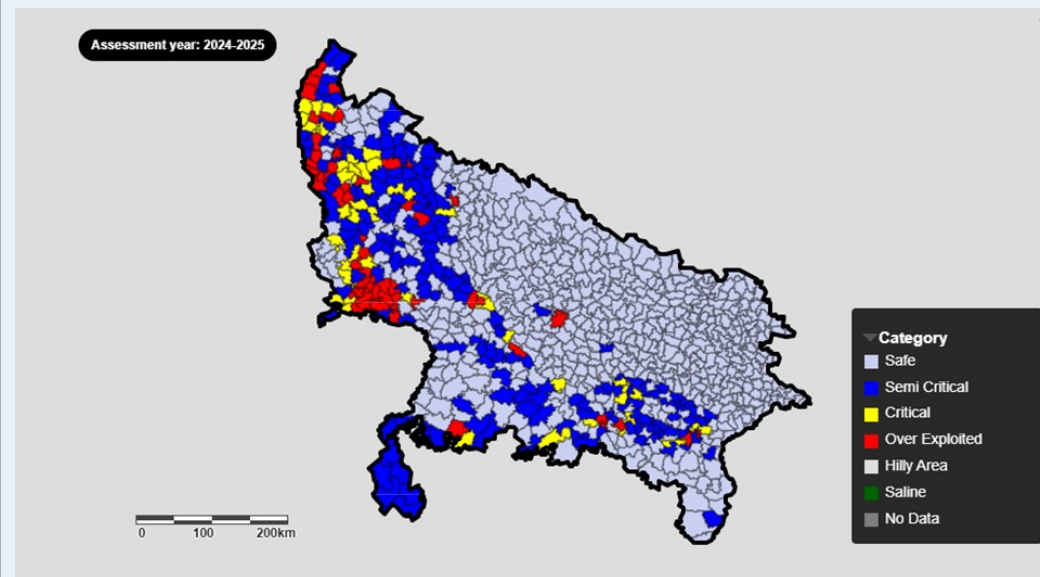
EC (3000 $\mu\text{S/cm}$)	Agra, Aligarh, Amethi, Etawah, Firozabad, G.B. Nagar, Ghaziabad, Ghazipur, Hathras, Mainpuri, Mathura
Fluoride ($F > 1.5$ mg/L)	Agra, Aligarh, Auraiya, Azamgarh, Farrukhabad, Fatehpur, Firozabad, G.B. Nagar, Ghazipur, Jhansi, Kannauj, Kanpur Dehat, Kanpur Nagar, Kaushambi, Lucknow, Mahoba, Mainpuri, Mathura, Meerut, Pratapgarh, Prayagraj, Raebareli, Sonbhadra, Unnao
Nitrate (Nitrate > 45 mg/L)	Agra, Aligarh, Amethi, Amroha, Auraiya, Baghpat, Balrampur, Banda, Bijnor, Budaun, Chitrakoot, Etah, Etawah, Fatehpur, Firozabad, G.B. Nagar, Ghaziabad, Ghazipur, Gorakhpur, Hamirpur, Hapur, Hathras, Jalaun, Jaunpur, Jhansi, Kanpur Dehat, Kanpur Nagar, Kasganj, Kaushambi, Lalitpur, Lucknow, Mahoba, Mainpuri, Mathura, Meerut, Mirzapur, Moradabad, Muzaffarnagar, Pilibhit, Prayagraj, Rampur, Sambhal, Sant Ravidas Nagar, Shamli, Siddharth Nagar, Sonbhadra, Unnao, Varanasi
Arsenic ($As > 10$ ppb)	Bahraich, Balrampur, Ballia, Barabanki, Bareilly, Bijnor, Farrukhabad, Fatehpur, Lakhimpur Kheri, Moradabad, Pilibhit, Rampur, Sambhal, Shahjahanpur
Uranium ($U > 30$ ppb)	Agra, Aligarh, Banda, Bijnor, Budaun, Etah, G.B. Nagar, Ghaziabad, Ghazipur, Jalaun, Jhansi, Kannauj, Kanpur Dehat, Lalitpur, Mahoba, Meerut, Prayagraj, Raebareli, Shahjahanpur, Sonbhadra, Unnao, Varanasi

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Central Ground Water Board Department of Water Resources, RD & GR Ministry of Jal Shakti, Government of India



Dynamic Ground Water Resources, Ground Water Level & Ground Water Quality of Uttar Pradesh, 2025

November, 2025

Background

- ◆ Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- ◆ Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- ◆ As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Block).
- ◆ The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- ◆ GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	886.75 mm
2	Hydrogeology	Five distinct hydrogeological units: 1. Bhabar, 2. Terai, 3. Central Ganga Plain-most promising and multiple aquifer system, 4. Marginal alluvial plain and 5. Southern hard rock terrain of Bundelkhand area.
3	Recharge Worthy Area of the State	2.30 Lakh Sq. Km
4	Assessment Unit (AU) Type / Number	Block / 836 Numbers
5	Average area of Assessment Unit	274.57 Sq. Km

Findings

	Attribute	GWRA-	GWRA-	GWRA-	GWRA-	GWRA-	GWRA
1	Total Annual Ground Water	69.92	72.2	71.45	71.83	72.84	73.39
2	Annual Extractable Ground	65.32	66.88	65.3	65.57	66.38	66.97
3	Annual Ground Water Ex-	45.84	46.03	46.14	46.4	46.76	46.89
4	Stage of Ground Water Ex- traction (in %)	70.18	68.83	70.66	70.76	70.45	70.00

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

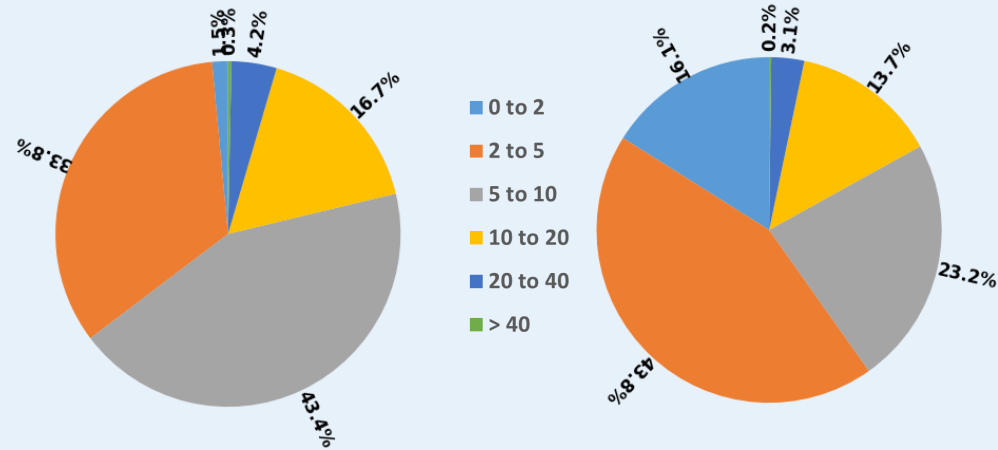
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Num- ber of AUs	% of AUs	Num- ber of AUs	% of AUs	Num- ber of AUs	% of AUs	Num- ber of AUs	% of AUs	Num- ber of AUs	% of AUs	Num- ber of AUs	% of AUs
1	Safe	540	65	541	65	557	67	559	66.87	566	67.7	563	67.34
2	Semi-critical	151	18	174	21	169	20	172	20.57	165	19.74	171	20.45
3	Critical	48	6	49	6	47	6	43	5.14	46	5.5	48	5.74
4	Over-exploited	91	11	66	8	63	8	62	7.42	59	7.06	54	6.46
5	Saline												
Total number of AUs		830		830		836		836		836		836	

Recommendations: -

- * The State of Uttar Pradesh is categorized with five distinct hydrogeological units – Bhabar, Terai, Central Ganga Plains, Marginal Alluvial Plain, Southern Hardrock area. Bhabar is mainly the recharge zone having deeper water levels. Terai zone lies between Bhabar in the North and Central Ganga Plain in the South. It is characterized by fine grained sediments with occasional pebbles and boulders. Central Ganga Plain constitutes the most promising ground water repository characterized by multi-layered aquifer systems. Southern part mainly occupied by Hard rocks comprising of Granite/ Granitic Gneiss and Marginal Alluvium in Bundelkhand Region and Vindyan Sedimentary formations in Mirzapur and Sonbhadra Districts.
- * Total Annual Ground Water Recharge of the state has been assessed as 73.39 bcm and Annual Extractable Ground Water Resource as 66.97 bcm. The Annual Ground Water Extraction is 46.89 bcm and the Stage of Ground Water Extraction of the State is 70.00%. Out of the 836 assessment units consisting of 826 blocks and 10 cities, 54 units (6.46 %) have been categorized as 'Over- exploited', 48 units (5.74 %) as 'Critical', 171 units (20.45 %) as 'Semi-critical' and 563 units (67.3 %) as 'Safe'. Promoting / Incentivizing change for change in cropping pattern from high water consuming crops like paddy to less water consuming crops in the western UP.
- * In the safe category areas of Uttar Pradesh, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70%.
- * Restoration / rejuvenation of all the existing tanks should be taken up with the view of accommodating the available surface run off and thus augmentation of the ground water resources by artificial recharge. Periodical maintenance of these tanks is to be ensured. he "Manual on Artificial Recharge Techniques for augmentation of ground water" prepared by CGWB may be used for planning. (<https://cgwb.gov.in/sites/default/files/MainLinks/Manual-Artificial-Recharge.pdf>).
- * National Aquifer Mapping & Management Programme (NAQUIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpnmi/>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management. (<https://cgwb.gov.in/cgwbpnmi/>).
- * Creating awareness (Mass Awareness Campaign for public and farmers, slideshows, display boards on water conservation, Water Management Training Programme for personnel related with water sector, painting/essay competition for school students etc.) regarding water conservation etc may be organized at appropriate level.
- * In Industrial areas of Uttar Pradesh, Disposal of industrial effluents, solid waste and urban sewerage should be disposed off safely after treatment, so that the phreatic aquifer does not get adversely polluted .
- * State may review their free/subsidized electricity policy to farmers (if applicable), bring suitable water pricing policy and may work further towards crop rotation/diversification/other initiatives to reduce overdependence on groundwater.
- * Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Uttar Pradesh

1. Depth to Groundwater Level

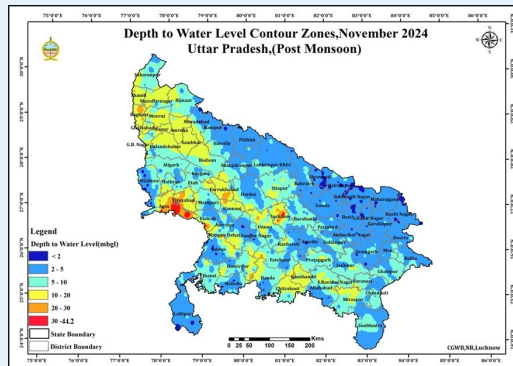
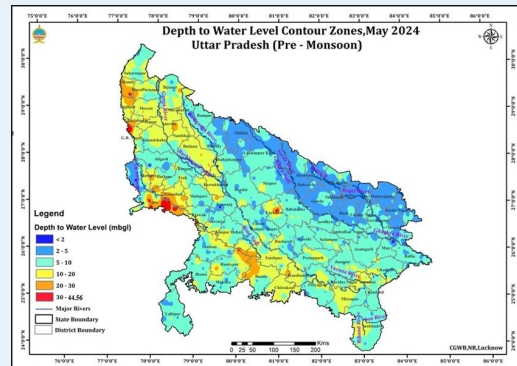


Pre-Monsoon (May 2024)

About 1.6% of wells recorded groundwater levels in 0–2 m bgl, 31.18% in 2–5 m bgl, 42.5% in 5–10 m bgl, 20.36% in 10–20 m bgl, 3.57% in 20–30 m bgl, and 0.79% in 30–40 m bgl, with overall levels ranging from 0.43 m to 44.56 m bgl, indicating generally shallow to moderately deep groundwater conditions with localized deep patches.

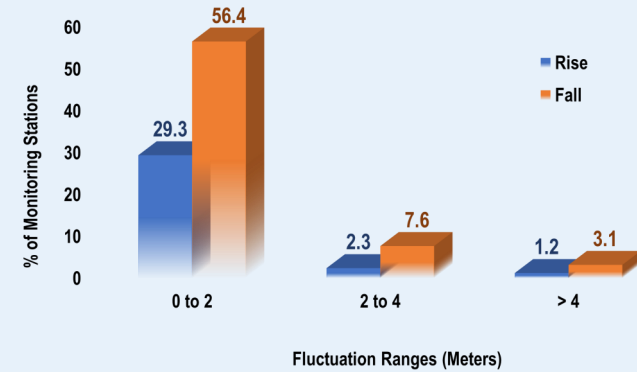
Post-Monsoon (November 2024)

About 14.86% of wells recorded groundwater levels in 0–2 m bgl, 44.59% in 2–5 m bgl, 22.25% in 5–10 m bgl, 14.41% in 10–20 m bgl, 3.15% in 20–30 m bgl, and 0.72% in 30–45 m bgl, with overall levels ranging from 0.38 m to 44.2 m bgl, indicating predominantly shallow to moderately



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

